

## bKash Recurring Payment Integration Guide [V 2.1.2]



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## Recurring Payment Platform

Merchant system will connect to bKash Recurring Payment platform via APIs. When a user initiates a subscription payment in their system, merchant will send request to bKash with the payment details. bKash will then provide a secure payment page where the customer will give consent for the entire subscription by providing and verifying their account information. Recurring Payment Platform will set up and maintain the payment subscription, and at the agreed cycle time, deduct the agreed amount from the customer's bKash account.

## Customer Interaction Flow

1. Select subscription at merchant website/app
2. Agree terms and conditions

bKash
Payment

according to subscription details i.e. start date, end date, fixed payment amount and payment cycle.

- You will receive a pre-notification SMS before payment date (except for daily subscription), you are requested to maintain sufficient balance on your bKash Account on such payment date.
- If the auto payment is successful, you will receive payment success notification through SMS.
- If the auto payment fails, you will be notified through SMS about such payment failure and bKash will retry the auto payment as per the agreed timeline between bKash and Merchant.
- If auto payment fails due to insufficient balance/account status, merchant may discontinue the service you are receiving.
- After cancellation of the subscription, you will be notified through SMS. You can re-subscribe as many times as you want.
- Through this process merchant is not receiving any confidential information related to your bKash Account.
- If you experience any problem to unsubscribe from Merchant's platform, please contact bKash customer support on 16247

Close
I Agree

16247

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3. Enter wallet number

bKash
Payment

Merchant: RPP Merchant UAT 50033  
Invoice: onetwo100

Amount: ₳ 50.00

Start Date: 2021-07-14

End Date: 2021-08-20

Cycle: 15 Days

Next Date: 2021-07-29

Next Amount: ₳ 50.00

Total No. of Payments: 4

Your bKash Account Number

e.g 01XXXXXXXXXX

Close

Next

16247

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4. Enter OTP

bKash
Payment

Merchant: RPP Merchant UAT 50033  
Invoice: onetwo100

Amount: ₳ 50.00

Start Date: 2021-07-14

End Date: 2021-08-20

Cycle: 15 Days

Next Date: 2021-07-29

Next Amount: ₳ 50.00

Total No. of Payments: 4

Enter Verification Code

bKash Verification Code

Didn't receive code?
Resend Code

Close

Next

16247

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## 5. Enter PIN

The screenshot shows the bKash Payment interface. At the top, there's a header with the bKash logo and 'Payment'. Below this, a merchant information box displays 'Merchant: RPP Merchant UAT 50033' and 'Invoice: onetwo100'. A table of payment details follows:

|                          |                        |
|--------------------------|------------------------|
| Amount: tk 50.00         | Start Date: 2021-07-14 |
| End Date: 2021-08-20     | Cycle: 15 Days         |
| Next Date: 2021-07-29    | Next Amount: tk 50.00  |
| Total No. of Payments: 4 |                        |

Below the table is a red box with the text 'Enter PIN' and a white input field labeled 'bKash PIN'. At the bottom of this box are 'Close' and 'Next' buttons. The footer includes the bKash logo, the number '16247', and the copyright notice '© 2021 All rights reserved by bKash'.

## 6. Complete the customer validation steps and redirect to merchant site/APP

The screenshot shows the bKash Payment interface after successful verification. The merchant information and payment details table are the same as in the previous step. Below the table, a red box displays the message 'Account successfully verified' and 'Please don't close the browser.' followed by 'Redirect in: 0'. The footer remains the same with the bKash logo, '16247', and the copyright notice.

## Subscription Creation Steps

1. A customer can sign up for a subscription in a merchant's platform (Website/ App)
2. Customer will see subscription packages including following information in merchant's platform
  - 
  - ✓ Payment cycle
  - ✓ Amount per payment cycle
  - ✓ Start Date
  - ✓ End date

### Note:

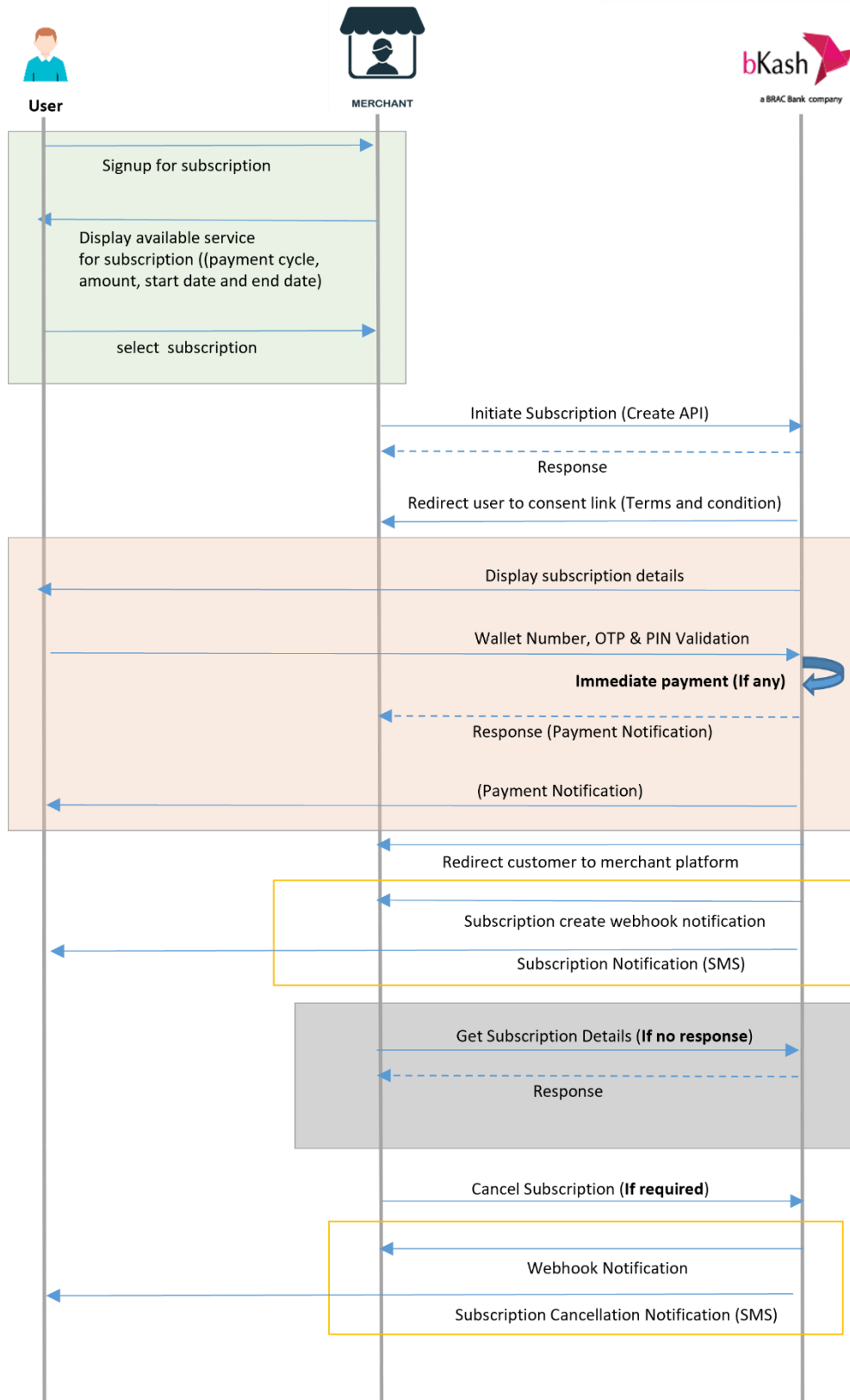
- *We are providing several payment cycles such as daily, 7 days, 15 days, 30 days, 90 days, 180 days, calendar month and calendar year. However, it will be the merchant who will determine which cycle to use for a particular subscription and will only display those particular cycle options.*
  - *Start date can be same day of subscription signup. If merchant is willing to provide the service right after the subscription, they can configure subscription sign up date as the start date of payment.*
  - *Merchant system will provide the end date of each payment subscription based on their agreement with their customer. End date can start from the following day after start date and be up to a maximum of 2 years from start date.*
3. In order to subscribe for a service, customer will first choose a subscription
  4. Next, merchant platform will initiate a subscription request to bKash via API
  5. bKash secure payment page will open within merchant platform UI and bKash system will redirect customer to a consent link.
  6. Customer will be able to view the terms & condition of creating a subscription first.

7. After agreeing to terms & conditions, the secure payment page will display the fixed criteria that the customer has agreed to the merchant page while proceeding - *Amount, Start Date, End Date, Cycle, Next Date, Next Amount and Total No. of Payments.*
8. Then one after one, customer will enter bKash wallet number, OTP and bKash PIN to give consent for the entire subscription so that bKash can debit that particular customer account during each cycle without needing repeated consent throughout the subscription period

Note:

*bKash system checks validation for customer as per following:*

- *Registered bKash customer account*
  - *Identity & account status 'Active'*
  - *No state tag is restricting debit transaction*
9. bKash system will set up subscription in its own subscription engine
  10. bKash will provide notification of subscription set up to both customer (through SMS) and merchant (through webhook).
  11. After successful subscription, customer will be redirected to merchant's platform i.e. website/ App
  12. bKash system will have a log of each subscription



Flow: System interaction flow between merchant and bKash

## Cycle payment:

### Payment Flow:

1. bKash system will send a notification before payment date as per the following schedule.

| Cycle    | Pre-Payment Notification Day |
|----------|------------------------------|
| Daily    | N/A                          |
| 7 Days   | 1                            |
| 15 Days  | 2                            |
| 30 Days  | 3                            |
| 90 Days  | 3                            |
| 180 Days | 3                            |
| Monthly  | 3                            |
| Yearly   | 3                            |

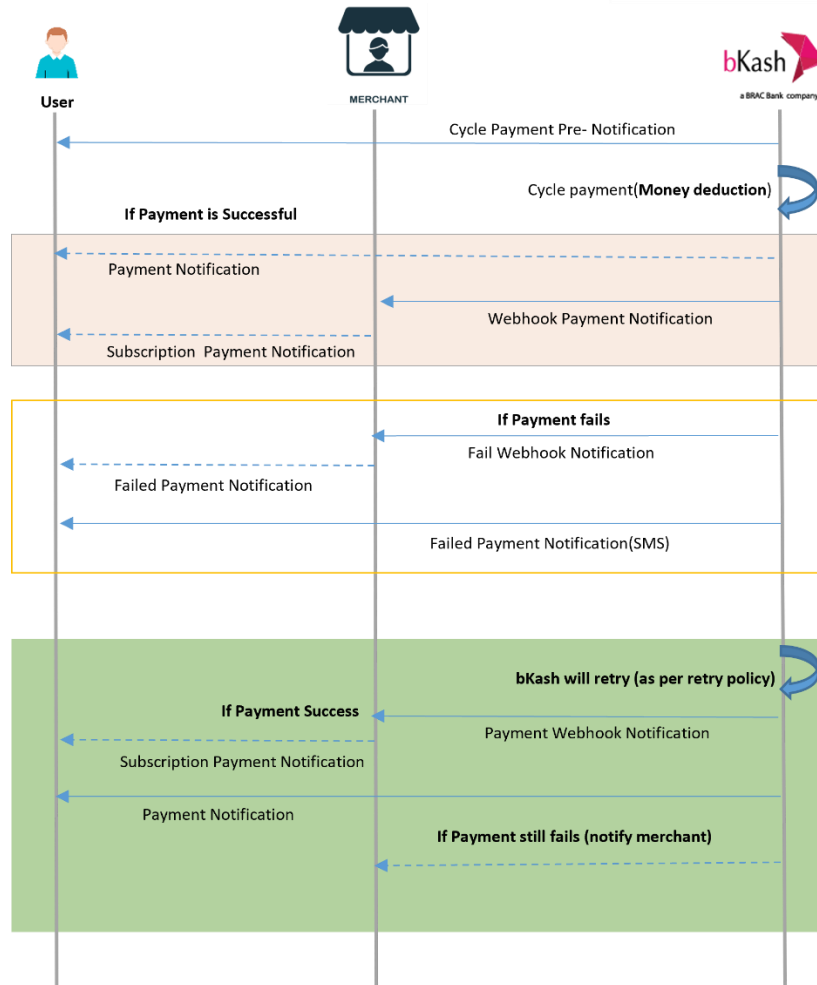
2. Payment will be automatically initiated by bKash system at each cycle. For example, if subscription was set up as monthly, with starting date of 5 June, then payment will automatically be made on 5th of each month.
3. If payment is successful, both customer and merchant will be sent successful notifications.
4. If payment fails, customer will be notified of payment failure through SMS and merchant will receive webhook notification.
5. In case of failed payment, bKash system will retry payment as per the following policy.

| Cycle  | Pre-notification day | No. of Retrial | Retrial Period        |
|--------|----------------------|----------------|-----------------------|
| Daily  | N/A                  | N/A            | N/A                   |
| 7 Days | 1                    | 1              | 2 Days After Pay Date |

|          |   |   |                             |
|----------|---|---|-----------------------------|
| 15 Days  | 2 | 2 | Every 3 Days After Pay Date |
| 30 Days  | 3 | 2 | Every 3 Days After Pay Date |
| 90 Days  | 3 | 2 | Every 3 Days After Pay Date |
| 180 Days | 3 | 2 | Every 3 Days After Pay Date |
| Monthly  | 3 | 2 | Every 3 Days After Pay Date |
| Yearly   | 3 | 2 | Every 3 Days After Pay Date |

**\*\* Merchant will get notification for each retrial attempt.**

- If payment still fails, merchant will need to manage the payment of that particular cycle with customer directly.
- bKash system will continue with regular subscription payment from the next cycle.



## Subscription Cancellation

### Customer Initiated

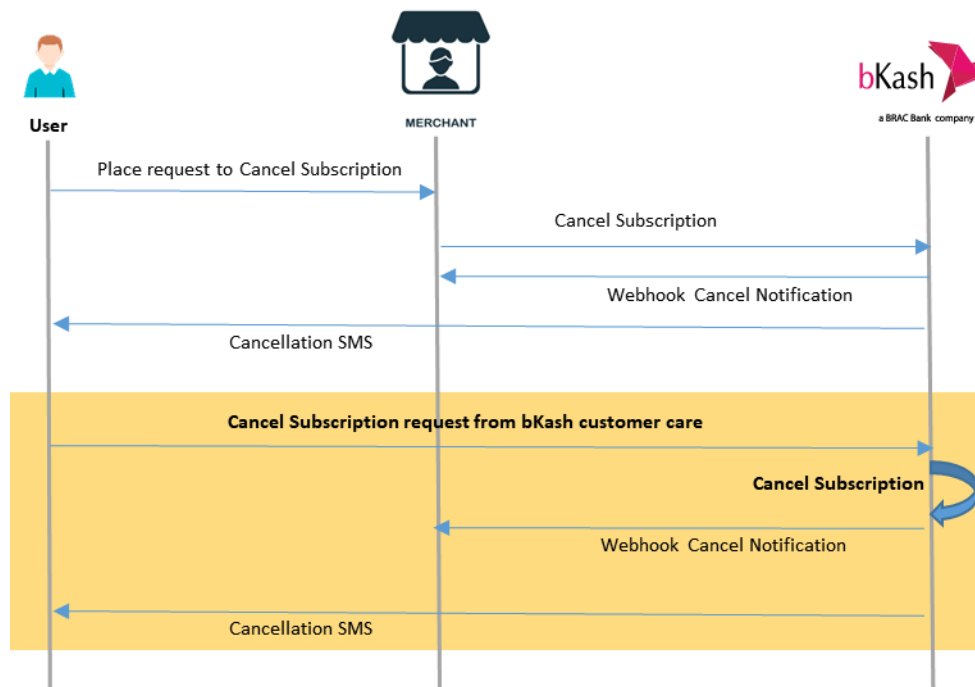
#### A. From Merchant's Website/App:

1. If a customer is willing to cancel any of his/her existing subscription, s/he will be able to do so on the merchant's website/App
2. User will log into the merchant's website/App, go the subscribed services and select 'Cancel Subscription'. This is an example of the most simplified process of cancellation for any subscription. This process might vary from merchant to merchant
3. After receiving the cancelation request, merchant will forward the request to bKash immediately
4. Once received, bKash system will execute the request automatically.

5. After successful cancellation, both merchant and customer

will be notified **B. From bKash Customer Service:**

- If a customer is willing to cancel any of his/her existing subscription, and for some reason cannot get it canceled through his/her merchant, s/he can do so by calling bKash customer care
- CS will validate the customer's identity in their usual process
- Upon necessary validation, CS will search by customer account number in CIMT portal
- CIMT System will show all the subscription under that customer account, and will show the subscription ID, start date, end date, amount, frequency, status (active/ended/canceled)
- CS will cancel subscription in CIMT



## Create Subscription

### Create

To initiate subscription, merchant can call create subscription API to communicate with bKash API. JSON data should be provided during this API call.

| METHOD | URL               |
|--------|-------------------|
| POST   | /api/subscription |

| TYPE      | Data Type | Mandatory/<br>Optional | PARAMETERS   | Sample Values                        | Description                                                                                             |
|-----------|-----------|------------------------|--------------|--------------------------------------|---------------------------------------------------------------------------------------------------------|
| HEAD      | String    | M                      | version      | v1.0                                 | Version Number indicates the interface version.                                                         |
| HEAD      | Number    | M                      | channelId    | Customer APP                         | Channel Id indicates channel by which user initiated the request.<br>1: Customer APP<br>2: Merchant WEB |
| HEAD      | Instant   | M                      | timeStamp    | 2021-02-08T09:16:28.603855Z          | UTC Instant                                                                                             |
| HEAD      | String    | M                      | x-api-key    |                                      | API Key                                                                                                 |
| HEAD      | String    | M                      | Content-Type | application/json                     | The Content-Type representation header is used to indicate the original media type of the resource      |
| POST BODY |           | M                      |              | <a href="#">Request Body (below)</a> | Part of a post request                                                                                  |

## Request Body

| Data Type | Parameter             | Mandatory /Optional | Sample Value                           | Remarks                       | Description                                                                                                                      |
|-----------|-----------------------|---------------------|----------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| String    | subscriptionRequestId | M                   | <1sagsadfa<br>ssddasdfsdf<br>fasdfgda> | UNIQUE VALUE                  | System generated unique id                                                                                                       |
| Number    | serviceId             | M                   | <1>                                    | Subject to updated over time. | Provided by bKash                                                                                                                |
| String    | redirectURL           | M                   | <string>                               | Mandatory for merchants       | Provided by merchant                                                                                                             |
| String    | paymentType           | M                   | [FIXED]                                |                               | Indicates recurring payment type                                                                                                 |
| String    | subscriptionType      | M                   | [BASIC]                                | Basic or With Payment         | BASIC - indicates only scheduled payment and<br>WITH_PAYMENT - indicates a mandatory immediate payment for subscription creation |
| Decimal   | amount                | M                   | <35>                                   |                               | Amount that will be recurrently paid                                                                                             |

|                |                             |   |           |                                                                                                                             |                                                                                     |
|----------------|-----------------------------|---|-----------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Decimal</b> | firstPaymentAmount          | O | <Decimal> | Mandatory when<br>WITH_PAYMENT                                                                                              | Amount needs to<br>be paid when<br>subscription is<br>created for the<br>first time |
| <b>String</b>  | currency                    | M | [BDT]     |                                                                                                                             | Currency of the<br>Payment                                                          |
| <b>Boolean</b> | firstPaymentIncludedInCycle | O | <Boolean> | Value – 0 [If this<br>payment is not<br>part of the<br>payment]<br>Value – 1 [If this<br>payment is part<br>of the payment] | If subscription<br>has first payment<br>during<br>registration                      |
| <b>Decimal</b> | maxCapAmount                | O | <5000>    |                                                                                                                             | Maximum value<br>of a payment                                                       |
| <b>String</b>  | frequency                   | M | <WEEKLY>  | Available cycles<br>for each<br>subscriptions                                                                               | DAILY<br>WEEKLY<br>FIFTEEN_DAYS<br>THIRTY_DAYS<br>NINETY_DAYS<br>ONE_EIGHTY_DA      |
|                |                             |   |           |                                                                                                                             | YS<br>CALENDAR_MON<br>TH<br>CALENDAR_YEAR                                           |

|            |                       |   |               |                                     |                                                                                         |
|------------|-----------------------|---|---------------|-------------------------------------|-----------------------------------------------------------------------------------------|
| String     | startDate             | M | 2021-04-10    | ISO 8601 Date Format                | Start date of a subscription. From this date, payment deduction will take place         |
| String     | expiryDate            | M | 2025-04-10    | ISO 8601 Date Format                | Expiry date of a subscription. After this date, no payment will take place.             |
| String     | merchantShortCode     | O | <Idlc>        | Unique Code for Merchant            | Provided by bKash                                                                       |
| String     | payerType             | M | <CUSTOMER>    |                                     | Indicates Payer Type whether it is Customer or any partner organization                 |
| String     | payer                 | M | <01870725511> |                                     | Wallet ID of a customer                                                                 |
| String     | subscriptionReference | M | <string>      |                                     | The subscription reference shared by merchant to                                        |
|            |                       |   |               |                                     | refer to the subscription service                                                       |
| Parameters | extraParams           | O |               | Multiple Data as key, value format. | Indicates the dynamic parameters. To capture the extra parameter for recurring payment. |

## Response

| Type   | Data Type      | Parameters            | Sample Values    | Remarks                                                                             |
|--------|----------------|-----------------------|------------------|-------------------------------------------------------------------------------------|
| ACCEPT |                | <b>Content-Type</b>   | application/json |                                                                                     |
| JSON   | <b>String</b>  | subscriptionRequestId | <string>         | Subscription session id from request                                                |
| JSON   | <b>String</b>  | redirectURL           | <string>         | Redirect the customer to the redirect_url to get user consent for this subscription |
| JSON   | <b>String</b>  | expirationTime        | string(\$date)   | Timestamp giving the expiration date of this subscription (if any)                  |
| JSON   | <b>Instant</b> | timeStamp             | <date>           | Timestamp for when the subscription period (if any) began                           |

## Response [Failed]: JSON

| Type   | Data Type | Parameters            | Sample Values    | Remarks                              |
|--------|-----------|-----------------------|------------------|--------------------------------------|
| ACCEPT |           | <b>Content-Type</b>   | application/json |                                      |
| JSON   | String    | subscriptionRequestId | <string>         | Subscription session id from request |
| JSON   | String    | errorCode             | <string>         | Indicate result code                 |
| JSON   | String    | errorDescription      | <string>         | Indicate result description          |

## RESPONSE EXAMPLE:

```
{
  "subscriptionRequestId": "MerchantPROD0014",
  "redirectURL": "https://intent.recurring.pay.bka.sh/intent/web/intent/BTCCXTJF",
  "expirationTime": "2021-07-11T04:29:29.801417Z",
  "timeStamp": "2021-07-11T04:19:30.067816Z"
}
```

## Webhook Notification

Notifications must be verified to make sure a notification is indeed from bKash and not from a malicious source.

Here you have a signature, payload, and api\_key and you want to verify the signature.

For example:

```
payload = '{"message":null,"timeStamp":"2021-04-19T10:08:22.267540Z","subscriptionRequestId":"573a44e6-5686-465a-868ce7571b0233b1","paymentId":73,"paymentStatus":"SUCCEEDED_PAYMENT","subscriptionId":40,"trxId":"8DJ9022ZD7","amount":51.00,"dueDate":"2021-04-19","nextPaymentDate":"2021-04-26","trxDate":"2021-04-19T16:08:22","firstPayment":true}'; signature
= "VL5Qow6bpQQnhHIYzf35rs5nssCGRc-Hih6l61ZC8jY=" api_key =
"3EbBE-L30-PEDU-KjkhxpEop8SYsanOa"
```

**Signature decryption Steps:** Verify the payloads that bKash sends to your webhook endpoints, following python examples can be applied:

- Decode the base64-encoded header X-Signature to get the signature *signature = base64.urlsafe\_b64decode(signature)*
- Decode the base64-encoded api\_key to get the key.  
*key = base64.urlsafe\_b64decode(api\_key)*
- Generate hmac digest (using sha256 hashing) of the payload using the key *digest = hmac.new(key=key, msg=payload, digestmod=hashlib.sha256).digest()*
- Compare the digest data with signature (*hmac.compare\_digest(digest, signature)*)

## Subscription Webhook Request

### Post Method

| TYPE         | Data Type | Mandatory/<br>Optional | Parameters | Sample Values                           | Remarks                                                 |
|--------------|-----------|------------------------|------------|-----------------------------------------|---------------------------------------------------------|
| HEAD         | String    | M                      | Signature  | <Hash value>                            | Encrypted, Signature should be match                    |
| HEAD         | String    | M                      | Type       | <Payment>                               | Possible values:<br>SUBSCRIPTION,<br>PAYMENT,<br>REFUND |
| POST<br>BODY |           | M                      |            | <a href="#">Request Body</a><br>(below) | Part of a post request                                  |

### Request Body

| Data Type | Parameter             | Mandatory<br>/Optional | Sample<br>Value    | Remarks         | Description                                                 |
|-----------|-----------------------|------------------------|--------------------|-----------------|-------------------------------------------------------------|
| String    | subscriptionRequestId | M                      | <<br>TraggSB00108> |                 | Provided by bKash                                           |
| Number    | subscriptionId        | M                      | <141>              | UNIQUE<br>VALUE | The unique identifier<br>for the associated<br>Subscription |

|         |                    |   |                |                                            |                                                                                                                                                                                                                                                                                                                       |
|---------|--------------------|---|----------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| String  | subscriptionStatus | M | < VERIFIED>    |                                            | INITIALIZED:<br>Subscription creation request received by MW<br>VERIFIED: IF requester is trusted, it will be VERIFIED instantly, if untrusted after pin verification<br>SUCCEEDED: Payment completed successfully<br>FAILED: Payment failed or pin verification failed<br>CANCELLED: Cancelled by requester or admin |
| Decimal | Amount             | O | [Decimal]      |                                            | Amount that will be recurrently paid.                                                                                                                                                                                                                                                                                 |
| String  | trxId              | O | <string>       | The unique identifier for the Transaction  | Provided by bKash                                                                                                                                                                                                                                                                                                     |
| String  | trxDate            | O | string(\$date) | Will be present when subscription creation | The date indicating when the Transaction was done                                                                                                                                                                                                                                                                     |
| String  | nextPaymentDate    | M | 2021-07-14     | ISO 8601 Date Format                       | Next payment date of the subscription period                                                                                                                                                                                                                                                                          |

|         |           |   |                            |                                          |                                                                                                                     |
|---------|-----------|---|----------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Instant | timeStamp | M | 2021-0714T11:24:28.677781Z | Timestamp when this request is generated | Timestamp indicating when the Transaction was created                                                               |
| String  | payer     | M | 01832353841                | Senders Wallet Number                    | The Wallet which will be deducted for subscription                                                                  |
| String  | frequency | M | CALENDAR_MONTH             | Available cycles for each subscriptions  | DAILY<br>WEEKLY<br>FIFTEEN_DAYS<br>THIRTY_DAYS<br>NINETY_DAYS<br>ONE_EIGHTY_DAYS<br>CALENDAR_MONTH<br>CALENDAR_YEAR |

Note: \* All Webhooks Request will be initiated from RPP \* Just send HTTP Code 200 or 202 if you successfully received it. Otherwise send relevant HTTP Error Code **Sample Payload:**

```
{
  "message": null,
  "timeStamp": "2021-07-14T11:24:28.677781Z",
  "subscriptionRequestId": "TraggSB00108",
  "subscriptionId": 141,
  "subscriptionStatus": "SUCCEEDED",
  "nextPaymentDate": "2021-07-14",
  "amount": null,
  "trxId": null,
  "trxDate": null,
  "payer": "01832353841",
  "frequency": "CALENDAR_MONTH"
}
```

## Subscription Payment Webhook Request

### Post Method

| TYPE         | Data Type | Mandatory/<br>Optional | Parameters | Sample Values                            | Remarks                                                 |
|--------------|-----------|------------------------|------------|------------------------------------------|---------------------------------------------------------|
| HEAD         | String    | M                      | Signature  | <Hash value>                             | Encrypted, Signature should be match                    |
| HEAD         | String    | M                      | Type       | <Payment>                                | Possible values:<br>SUBSCRIPTION,<br>PAYMENT,<br>REFUND |
| POST<br>BODY |           | M                      |            | <a href="#">Request Body<br/>(below)</a> | Part of a post request                                  |

### Request Body

| Data Type | Parameter             | Mandatory<br>/Optional | Sample<br>Value | Remarks         | Description                              |
|-----------|-----------------------|------------------------|-----------------|-----------------|------------------------------------------|
| String    | subscriptionRequestId | M                      | <TraggSB00107>  |                 | Provided by bKash                        |
| Number    | paymentId             | M                      | <438>           | UNIQUE<br>VALUE | The unique identifier<br>for the payment |

|        |               |   |               |  |                                                                              |
|--------|---------------|---|---------------|--|------------------------------------------------------------------------------|
| String | paymentStatus | M | <INITIALIZED> |  | Possible values:<br>INITIALIZED,<br>PROCESSING_PAYMENT,<br>PROCESSING_REFUND |
|--------|---------------|---|---------------|--|------------------------------------------------------------------------------|

|         |                |   |           |                 |                                                                                                                                                                                                                               |
|---------|----------------|---|-----------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |                |   |           |                 | ,<br>SUCCEEDED_PAYMENT,<br>SUCCEEDED_REFUND,<br>FAILED_PAYMENT,<br>FAILED_REFUND,<br>RE_SUCCEEDED_PAYMENT,<br>RE_SUCCEEDED_REFUND,<br>RE_FAILED_PAYMENT,<br>RE_FAILED_REFUND,<br>RE_NOT_FOUND_PAYMENT,<br>RE_NOT_FOUND_REFUND |
| Boolean | firstPayment   | M | <Boolean> |                 | If subscription has first payment during registration                                                                                                                                                                         |
| Number  | subscriptionId | M | <140>     | UNIQUE<br>VALUE | The unique identifier for the associated Subscription                                                                                                                                                                         |

|         |                 |   |                            |                                            |                                                       |
|---------|-----------------|---|----------------------------|--------------------------------------------|-------------------------------------------------------|
| Decimal | amount          | M | <30>                       | Payable amount per cycle                   | Amount that will be recurrently paid.                 |
| String  | trxId           | M | <8GC004U54M>               | The unique identifier for the Transaction  | Provided by bKash                                     |
| String  | dueDate         | M | 2021-07-12                 | ISO 8601 Date Format                       | Due date when payment supposed to deduct              |
| String  | trxDate         | M | 2021-07-12T05:16:18        | Will be present when subscription creation | The date indicating when the Transaction was done     |
| String  | nextPaymentDate | M | 2021-07-13                 | ISO 8601 Date Format                       | Next payment date of the subscription period          |
| Instant | timeStamp       | M | 2021-0712T05:16:23.905207Z | Timestamp when this request is generated   | Timestamp indicating when the Transaction was created |
| String  | payer           | M | 01832353841                | Senders Wallet Number                      | The Wallet which will be deducted for subscription    |

|        |           |   |                |                                         |                                                                                                                           |
|--------|-----------|---|----------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| String | frequency | M | CALENDAR_MONTH | Available cycles for each subscriptions | DAILY<br>WEEKLY<br>FIFTEEN_DAYS<br>THIRTY_DAYS<br>NINETY_DAYS<br>ONE_EIGHTY_DAYS<br>CALENDAR_MONTH<br>TH<br>CALENDAR_YEAR |
|--------|-----------|---|----------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------|

#### Sample Payload:

```
{
  "message": null,
  "timeStamp": "2021-07-12T05:16:23.905207Z",
  "subscriptionRequestId": "TraggSB00107",
  "paymentId": 438,
  "paymentStatus": "SUCCEEDED_PAYMENT",
  "subscriptionId": 140,
  "trxId": "8GC004U54M",
  "amount": 1,
  "dueDate": "2021-07-12",
  "nextPaymentDate": "2021-07-13",
  "trxDate": "2021-07-12T05:16:18",
  "firstPayment": false
  "payer": "01832353841",
  "frequency": "CALENDAR_MONTH"
}
```

## Refund Webhook Request

### Post Method

| TYPE         | Data Type | Mandatory/<br>Optional | Parameters | Sample Values                           | Remarks                                                 |
|--------------|-----------|------------------------|------------|-----------------------------------------|---------------------------------------------------------|
| HEAD         | String    | M                      | Signature  | <Hash value>                            | Encrypted, Signature should be match                    |
| HEAD         | String    | M                      | Type       | < REFUND>                               | Possible values:<br>SUBSCRIPTION,<br>PAYMENT,<br>REFUND |
| POST<br>BODY |           | M                      |            | <a href="#">Request Body</a><br>(Below) | Part of a post request                                  |

### Refund Webhook Request Body

| Data Type | Parameter             | Mandatory<br>/Optional | Sample<br>Value | Remarks                                                                    | Description                                              |
|-----------|-----------------------|------------------------|-----------------|----------------------------------------------------------------------------|----------------------------------------------------------|
| String    | subscriptionRequestId | M                      | <TraggSB00108>  |                                                                            | Provided by bKash                                        |
| Number    | paymentId             | M                      | <444>           | UNIQUE<br>VALUE                                                            | The unique identifier for the payment                    |
| String    | paymentStatus         | M                      | <INITIALIZED>   | Possible values:<br>INITIALIZED,<br>PROCESSING_<br>PAYMENT,<br>PROCESSING_ | INITIALIZED – means payment is still pending to process. |

|                |                |   |           |                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                 |
|----------------|----------------|---|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                |                |   |           | REFUND,<br>SUCCEEDED_P<br>AYMENT,<br>SUCCEEDED_R<br>EFUND,<br>FAILED_PAYM<br>ENT,<br>FAILED_REFU<br>ND,<br>RE_SUCCEDE<br>D_PAYMENT,<br>RE_SUCCEDE<br>D_REFUND,<br>RE_FAILED_PA<br>YMENT,<br>RE_FAILED_RE<br>FUND,<br>RE_NOT_FOU<br>ND_PAYMENT<br>,<br>RE_NOT_FOU<br>ND_REFUND | PROCESSING_PAYME<br>NT – Payment<br>processing is ongoing.<br><br>PROCESSING_REFUND<br>– Refund process is<br>ongoing.<br><br>SUCCEEDED_PAYMEN<br>T – Payment has been<br>deducted.<br><br>SUCCEEDED_REFUND<br>– Refund done<br>successfully.<br><br>FAILED_PAYMENT –<br>Payment failed after<br>retry.<br><br>FAILED_REFUND –<br>Refund failed |
| <b>Boolean</b> | firstPayment   | M | <Boolean> |                                                                                                                                                                                                                                                                               | If subscription has first<br>payment during<br>registration                                                                                                                                                                                                                                                                                     |
| <b>Number</b>  | subscriptionId | M | <141>     | UNIQUE<br>VALUE                                                                                                                                                                                                                                                               | The unique identifier<br>for the associated<br>Subscription                                                                                                                                                                                                                                                                                     |

|         |                |   |                            |                                          |                                                                                                                     |
|---------|----------------|---|----------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| String  | reverseTrxId   | M | <8GE704UB7F >              | Refund transaction ID (If any refund)    | Provided by bKash                                                                                                   |
| Decimal | refundedAmount | M | <10>                       |                                          | The refunded amount merchant has requested (If any)                                                                 |
| String  | reverseTrxDate | M | 2021-07-14T11:24:28        |                                          | Refund transaction date (If any refund)                                                                             |
| Instant | Timestamp      | M | 2021-0714T11:25:54.426741Z | Timestamp when this request is generated | Timestamp indicating when the Transaction was created                                                               |
| String  | payer          | M | 01832353841                | Senders Wallet Number                    | The Wallet which will be deducted for subscription                                                                  |
| String  | frequency      | M | CALENDAR_MONTH             | Available cycles for each subscriptions  | DAILY<br>WEEKLY<br>FIFTEEN_DAYS<br>THIRTY_DAYS<br>NINETY_DAYS<br>ONE_EIGHTY_DAYS<br>CALENDAR_MONTH<br>CALENDAR_YEAR |

## Sample Payload:

```
{
  "message": null,
  "timeStamp": "2021-07-14T11:25:54.426741Z",
  "subscriptionRequestId": "TraggSB00108",
  "paymentId": 444,
  "paymentStatus": "SUCCEEDED_REFUND",
  "subscriptionId": 141,
  "reverseTrxId": "8GE704UB7F",
  "refundedAmount": 1,
  "reversTrxDate": "2021-07-14T11:24:28",
  "firstPayment": false
  "payer": "01832353841",
  "frequency": "CALENDAR_MONTH"
}
```

## Cancel Subscription Webhook Request

### Post Method

| TYPE         | Data Type | Mandatory/<br>Optional | Parameters | Sample Values                        | Remarks                                                 |
|--------------|-----------|------------------------|------------|--------------------------------------|---------------------------------------------------------|
| HEAD         | String    | M                      | signature  | <Hash value>                         | Encrypted, Signature should be match                    |
| HEAD         | String    | M                      | type       | <Payment>                            | Possible values:<br>SUBSCRIPTION,<br>PAYMENT,<br>REFUND |
| POST<br>BODY |           | M                      |            | <a href="#">Request Body (below)</a> | Part of a post request                                  |

## Cancel Webhook Request Body

| Data Type | Parameter             | Mandatory /Optional | Sample Value                  | Remarks                                  | Description                                                                      |
|-----------|-----------------------|---------------------|-------------------------------|------------------------------------------|----------------------------------------------------------------------------------|
| String    | subscriptionRequestId | M                   | < TraggSB00108>               |                                          | Provided by bKash                                                                |
| Number    | subscriptionId        | M                   | <141>                         | UNIQUE VALUE                             | The unique identifier for the associated Subscription                            |
| String    | subscriptionStatus    | M                   | <CANCEELED>                   |                                          | Subscription status                                                              |
| String    | cancelledBy           | M                   | < Requestor>                  |                                          | Requestor –when cancelled by Merchant                                            |
| Instant   | timestamp             | M                   | <2021-0714T11:27:34.462238Z > | Timestamp when this request is generated | Timestamp indicating when the Transaction was created                            |
| String    | payer                 | M                   | 01832353841                   | Senders Wallet Number                    | The Wallet which will be deducted for subscription                               |
| String    | frequency             | M                   | CALENDAR_MONTH                | Available cycles for each subscriptions  | DAILY<br>WEEKLY<br>FIFTEEN_DAYS<br>THIRTY_DAYS<br>NINETY_DAYS<br>ONE_EIGHTY_DAYS |

|  |  |  |  |  |                                     |
|--|--|--|--|--|-------------------------------------|
|  |  |  |  |  | CALENDAR_MON<br>TH<br>CALENDAR_YEAR |
|--|--|--|--|--|-------------------------------------|

#### Sample Payload:

```
{
  "message": null,
  "timeStamp": "2021-07-14T11:27:34.462238Z",
  "subscriptionRequestId": "TraggSB00108",
  "subscriptionId": 141,
  "subscriptionStatus": "CANCELLED",
  "cancelledBy": "Requester",
  "payer": "01832353841",
  "frequency": "CALENDAR_MONTH"
}
```

#### Expiry Webhook Response Body

This web hook notification will be initiated from the Recurring Platform. For any subscription reaches the “END TIME”, the platform will allocate the subscription and trigger the web hook response to the designated aggregator.

The parameter details and sample payload are provided below:

| Data Type | Parameter             | Mandatory /Optional | Sample Value                  | Remarks                              | Description                                                            |
|-----------|-----------------------|---------------------|-------------------------------|--------------------------------------|------------------------------------------------------------------------|
| String    | Message               | M                   | <Null>                        |                                      | Provided by bKash                                                      |
| Instant   | Time Stamp            | M                   | <2023-02-07T06:16:52.084708Z> | Timestamp when this Web hook is sent | Event Time & Date of the Subscription Expiration web hook Sent from MW |
| String    | SubscriptionRequestId | M                   | <AAA102>                      | System generated UID                 | Special Identifier of the subscription request                         |

|        |                      |   |                        |                                        |                                                                                            |
|--------|----------------------|---|------------------------|----------------------------------------|--------------------------------------------------------------------------------------------|
| Number | subscriptionId       | M | <12>                   | Unique Value                           | Unique Identifier of the Associated subscription                                           |
| String | subscriptionStatus   | M | <EXPIRED>              | Subscription Status                    | Status of Subscription after Expiration                                                    |
| String | lastSucceededPayment | M | <2022-12-29 >          | Last Successful Payment Deduction Date | Date of Last Successful payment Deducted from the user before the subscription was EXPIRED |
| String | Frequency            | M | < CALENDAR_MONTH>      | Subscription Cycle                     | Type of Subscription Deduction CYCLE                                                       |
| String | creationTimestamp    | M | <2022-11-29T05:14:16Z> | Subscription Creation Time Stamp       | The time & date when the subscription was created in RPP                                   |
| Number | payer                | M | <01832353841>          | Wallet number                          | The wallet from where the amount was deducted                                              |

#### Response Payload Sample:

```
{
  "message": null,
  "timeStamp": "2023-02-07T06:16:52.084708Z",
  "subscriptionRequestId": "AAA102",
  "subscriptionId": 72,
  "subscriptionStatus": "EXPIRED",
  "lastSucceededPayment": "2022-12-29",
  "frequency": "CALENDAR_MONTH",
  "creationTimestamp": "2022-11-29T05:14:16Z"
  "payer": "01832353841"
}
```

## Payment Reminder (Pre-Notification) Web hook Response Body

The Payment Reminder option was a pre-existing feature which was sent through SMS. Now we are introducing this web hook notification. For any subscription that reaches the “Payment Date”, based on configuration of Pre-Notification, the platform will allocate the subscription and trigger the web hook response to the designated aggregator.

Currently the web hook facility can be availed besides the SMS triggering facility. **Designated aggregator can have the option to choose the Web hook or the SMS or both.**

The parameter details and sample payload are provided below:

| Data Type | Parameter             | Mandatory /Optional | Sample Value                       | Remarks                              | Description                                                            |
|-----------|-----------------------|---------------------|------------------------------------|--------------------------------------|------------------------------------------------------------------------|
| String    | Message               | M                   | <Null>                             |                                      | Provided by bKash                                                      |
| Instant   | Time Stamp            | M                   | <2023-02-07T06:16:52.084708Z>      | Timestamp when this Web hook is sent | Event Time & Date of the Subscription Expiration web hook Sent from MW |
| String    | RequesterId           | M                   | < 170 >                            | System generated UID                 | Special Identifier of the subscription request                         |
| String    | subscriptionRequestId | M                   | <099ecd0cd4e64793b7399b07ad288d1c> | Unique Value                         | Unique Identifier of the Associated subscription                       |
| String    | subscriptionId        | M                   | <6092>                             | Unique Value                         | Unique Identifier of the Associated subscription                       |
| String    | subscriptionStatus    | M                   | <SUCCEEDED>                        | Subscription Status                  | Status of Subscription                                                 |
| String    | nextPaymentDate       | M                   | <2022-12-29 >                      | Upcoming date of Payment             | Date of NEXT payment to be Deducted based on Frequency                 |

|        |           |   |                  |                    |                                                      |
|--------|-----------|---|------------------|--------------------|------------------------------------------------------|
| Number | Payer     | M | <01832353841>    | Wallet number      | The wallet number which the amount would be deducted |
| String | Frequency | M | <CALENDAR_MONTH> | Subscription Cycle | Type of Subscription Deduction CYCLE                 |
| Number | amount    | M | <5>              | Deducted amount    | Amount which was deducted on each subscription       |

### Response Payload Sample

```
{
  "message": null,
  "timeStamp": "2023-02-16T05:00:04.475985Z",
  "requesterId": 170,
  "subscriptionRequestId": "099ecd0cd4e64793b7399b07ad288d1c",
  "subscriptionId": 6092,
  "subscriptionStatus": "SUCCEEDED",
  "nextPaymentDate": "2023-02-17",
  "payer": "01847263866",
  "frequency": "CALENDAR_MONTH",
  "amount": 5
}
```

## Subscription Query (Subscription Get Request)

### Subscription Query using subscription ID

Merchant can also call for details of a single subscription by its ID to get latest status.

| METHOD | URL                                 |
|--------|-------------------------------------|
| GET    | /api/subscriptions/<subscriptionId> |

| TYPE | Data Type | PARAMETERS | Sample Value | Remarks                                                                                                                               |
|------|-----------|------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------|
| HEAD | String    | Version    | V.01         | The version number is used to control the system version of both parties. In case one party upgrades the system, the other party will |

|             |         |                |                  |                                                                                                       |
|-------------|---------|----------------|------------------|-------------------------------------------------------------------------------------------------------|
|             |         |                |                  | Recognize it via the version No.to avoid abnormal transaction.                                        |
| <b>HEAD</b> | Number  | channelId      | <Customer APP>   | Initiation channel of user requests.<br>Subject to updated with time.<br>Customer APP<br>Merchant WEB |
| <b>HEAD</b> | Instant | timestamp      |                  | Will be used to combat replay attack                                                                  |
| <b>HEAD</b> | String  | Content-Type   | application/json | Content Type of request parameters                                                                    |
| <b>HEAD</b> | String  | X-API-KEY      | <string>         | API key to maintain authentication between bKash and merchant servers.                                |
| <b>PATH</b> |         | subscriptionId | integer          | Subscription ID fetched from execute subscription API.                                                |

Response: JSON

| Data Type     | Parameters            | Sample Values  | Remarks                                                         |
|---------------|-----------------------|----------------|-----------------------------------------------------------------|
| <b>Number</b> | Id                    | <105>          | Encapsulated in data params                                     |
| <b>Number</b> | requesterId           |                | Id of merchant                                                  |
| <b>String</b> | subscriptionRequestId | < TraggPROD03> | System generated unique id                                      |
| <b>String</b> | Status                | <INITIALIZED>  | INITIALIZED,<br>VERIFIED,<br>SUCCEEDED,<br>FAILED,<br>CANCELLED |

|         |                    |                     |                                                                                                                                                                              |
|---------|--------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| String  | createdAt          | 2021-07-11 05:23:23 | Indicates subscription creation date                                                                                                                                         |
| String  | modifiedAt         |                     | TimeStamp                                                                                                                                                                    |
| Number  | Serviced           | 100001              | bKash provided                                                                                                                                                               |
| String  | redirectURL        |                     | Dynamic part of the URL. Fixed part will be predefined & pre shared. After payer authentication Recurring Payment middleware will redirect the customer to the callback URL. |
| String  | paymentType        | <FIXED>             | Indicates recurring payment type                                                                                                                                             |
| String  | subscriptionType   | <BASIC>             | Indicates which type of subscription:<br>BASIC<br>WITH_PAYMENT                                                                                                               |
| Decimal | amount             | <decimal>           | Amount that will be recurrently paid                                                                                                                                         |
| Decimal | firstPaymentAmount | <decimal>           | First payment amount                                                                                                                                                         |
| String  | Frequency          | <WEEKLY>            | DAILY<br>WEEKLY<br>FIFTEEN_DAYS<br>THIRTY_DAYS<br>NINETY_DAYS                                                                                                                |

|            |                       |                |                                                            |
|------------|-----------------------|----------------|------------------------------------------------------------|
|            |                       |                | ONE_EIGHTY_DAYS<br>CALENDAR_MONTH<br>CALENDAR_YEAR         |
| String     | startDate             | String(\$date) | ISO 8601 Date Format                                       |
| String     | expiryDate            | String(\$date) | ISO 8601 Date Format                                       |
| String     | merchantId            |                | Unique Code for Merchant<br>Provided by bKash              |
| String     | payerType             | <CUSTOMER>     |                                                            |
| String     | Payer                 | <string>       | Wallet ID of the payer                                     |
| String     | Currency              | <BDT>          |                                                            |
| String     | subscriptionReference | < TAGG >       | Payment Unique Reference                                   |
| Parameters | extraParams           | <null>         | Multiple Data as key, value format.                        |
| String     | cancelledBy           | <null>         | Contains name of the entity who cancelled the subscription |
| String     | cancelledTime         | <null>         | Timestamp of cancel event                                  |
| Boolean    | Expired               | <false>        | Denotes subscriptions expiration state                     |
| Boolean    | Enabled               | <true>         | Denotes that subscription is active, i.e. in enabled state |

|         |                 |                                        |                                |
|---------|-----------------|----------------------------------------|--------------------------------|
| String  | Rrule           | <FREQ=DAILY;INTERVAL=1;UNTIL=20210712> | recurrent rule of subscription |
| Boolean | Active          | [true]                                 | enabled and not expired        |
| String  | nextPaymentDate | 2021-07-12                             | ISO 8601 Date Format           |

#### RESPONSE EXAMPLE:

```
{
  "id": 105,
  "createdAt": "2021-07-11 05:23:23",
  "modifiedAt": "2021-07-11 05:23:27",
  "subscriptionRequestId": "TraggPROD03",
  "requesterId": 5,
  "serviceId": 100001,
  "paymentType": "FIXED",
  "subscriptionType": "WITH_PAYMENT",
  "amountQueryUrl": null,
  "amount": 1.00,
  "firstPaymentAmount": 2.00,
  "maxCapRequired": false,
  "maxCapAmount": null,
  "frequency": "DAILY",
  "startDate": "2021-07-11",
  "expiryDate": "2021-07-12",
  "merchantId": 3,
  "payerType": "CUSTOMER",
  "payer": "01844165554",
  "currency": "BDT",
  "nextPaymentDate": "2021-07-12", "status":
  "SUCCEEDED",
  "subscriptionReference": "TAGG",
  "extraParams": null,
  "cancelledBy": null,
  "cancelledTime": null,
  "expired": false, "enabled": true,

  "rrule": "FREQ=DAILY;INTERVAL=1;UNTIL=20210712",
  "active": true
}
```

## Subscription List (Subscription List Request)

Merchant can also call for details of a list of subscription

| METHOD | URL                              |
|--------|----------------------------------|
| GET    | /api/subscriptions/{page}/{size} |

| TYPE | Data Type | PARAMETERS   | Sample Value                         | Remarks                                                                                                                                                                                              |
|------|-----------|--------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HEAD | String    | Version      | V.01                                 | The version number is used to control the system version of both parties. In case one party upgrades the system, the other party will recognize it via the version No.to avoid abnormal transaction. |
| HEAD | Number    | channelId    | <Merchant WEB>                       | Initiation channel of user requests.<br>Subject to updated with time.<br>Customer APP<br>Merchant WEB                                                                                                |
| HEAD | Instant   | timestamp    |                                      | Will be used to combat replay attack                                                                                                                                                                 |
| HEAD | String    | Content-Type | application/json                     | Content Type of request parameters                                                                                                                                                                   |
| HEAD | String    | X-API-KEY    | <string>                             | API key to maintain authentication between bKash and merchant servers.                                                                                                                               |
| Body |           |              | <a href="#">Request Body (below)</a> | Part of a post request                                                                                                                                                                               |

### Subscription List Request Body

| Data Type | Parameter | Mandatory /Optional | Sample Value | Remarks | Description      |
|-----------|-----------|---------------------|--------------|---------|------------------|
| Number    | Page      | M                   | < 2>         |         | Number of pages  |
| Number    | size      | M                   | <50>         |         | Size of the page |

Response: JSON

| Data Type                    | Parameters    | Sample Values                                                                                                                                                                                                                                                                                                                            | Remarks                                   |
|------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Number                       | size          | <5>                                                                                                                                                                                                                                                                                                                                      | Size of the page                          |
| Number                       | totalElements | <59>                                                                                                                                                                                                                                                                                                                                     | Total number of subscriptions             |
| Number                       | totalPages    | < 12>                                                                                                                                                                                                                                                                                                                                    | Total number of pages with the given size |
| Number                       | currentPage   |                                                                                                                                                                                                                                                                                                                                          |                                           |
| List of subscription details | content       | {<br>"id": 1144,<br>"subscriptionRequestId":<br>"NonTragg0191",<br>"merchantId": 9,<br>"merchantShortCode": null,<br>"payer": "01865200294",<br>"amount": 50.00,<br>"startDate": "2021-07-30",<br>"expiryDate": "2021-08-20",<br>"frequency": "WEEKLY",<br>"status": "SUCCEEDED",<br>"cancelledBy": null,<br>"cancelledTime": null<br>}, | All subscriptions                         |

## RESPONSE EXAMPLE:

```
{
  "size": 5,
  "totalElements": 59,
  "totalPages": 12,
  "content": [
    {
      "id": 1144,
      "subscriptionRequestId": "NonTragg0191",
      "merchantId": 9,
      "merchantShortCode": null,
      "payer": "01865200294",
      "amount": 50.00,
      "startDate": "2021-07-30",
      "expiryDate": "2021-08-20",
      "frequency": "WEEKLY",
      "status": "SUCCEEDED",
      "cancelledBy": null,
      "cancelledTime": null
    },
    {
      "id": 1141,
      "subscriptionRequestId": "NonTragg0190",
      "merchantId": 9,
      "merchantShortCode": null,
      "payer": "01798266040",
      "amount": 50.00,
      "startDate": "2021-07-30",
      "expiryDate": "2021-08-02",
      "frequency": "DAILY",
      "status": "SUCCEEDED",
      "cancelledBy": null,
      "cancelledTime": null
    },
    {
      "id": 1140,
```

```
"subscriptionRequestId": "NonTragg0151",
"merchantId": 9,
"merchantShortCode": null,
"payer": "01798266040",
"amount": 10.00,
"startDate": "2021-07-28",
"expiryDate": "2021-07-30",
"frequency": "DAILY",
"status": "FAILED",
"cancelledBy": null,
"cancelledTime": null
},
{
"id": 1139,
"subscriptionRequestId": "NonTragg0150",
"merchantId": 9,
"merchantShortCode": null,
"payer": "01798266040",
"amount": 10.00,
"startDate": "2021-07-28",
"expiryDate": "2021-07-30",
"frequency": "DAILY",
"status": "CANCELLED",
"cancelledBy": "REQUESTER",
"cancelledTime": "2021-07-28T11:47:28.461139Z"
},
{
"id": 1060,
"subscriptionRequestId": "NonTragg0511",
"merchantId": 9,
"merchantShortCode": null,
"payer": "01798266040",
"amount": 4.00,
"startDate": "2021-07-26",
"expiryDate": "2021-07-29",
"frequency": "DAILY",
"status": "CANCELLED",
"cancelledBy": "REQUESTER",
"cancelledTime": "2021-07-26T11:32:23.203106Z"
}
],
```

```
"currentPage": 0
}
```

## Payment List (Payment List Request)

Merchant can also call for details of a list of payment

| METHOD | URL                                                         |
|--------|-------------------------------------------------------------|
| GET    | /api/subscription/payment/bySubscriptionId/{subscriptionId} |

| TYPE | Data Type | PARAMETERS   | Sample Value                         | Remarks                                                                                                                                                                                              |
|------|-----------|--------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HEAD | String    | Version      | V.01                                 | The version number is used to control the system version of both parties. In case one party upgrades the system, the other party will recognize it via the version No.to avoid abnormal transaction. |
| HEAD | Number    | channelId    | <Merchant WEB>                       | Initiation channel of user requests.<br>Subject to updated with time.<br>Customer APP<br>Merchant WEB                                                                                                |
| HEAD | Instant   | timestamp    |                                      | Will be used to combat replay attack                                                                                                                                                                 |
| HEAD | String    | Content-Type | application/json                     | Content Type of request parameters                                                                                                                                                                   |
| HEAD | String    | X-API-KEY    | <string>                             | API key to maintain authentication between bKash and merchant servers.                                                                                                                               |
| Body |           |              | <a href="#">Request Body (below)</a> | Part of a post request                                                                                                                                                                               |

## Payment List Request Body

| Data Type | Parameter      | Mandatory /Optional | Sample Value | Remarks      | Description                                           |
|-----------|----------------|---------------------|--------------|--------------|-------------------------------------------------------|
| Number    | subscriptionId | M                   | <141>        | UNIQUE VALUE | The unique identifier for the associated Subscription |

## Response: JSON

| Data Type               | Parameters | Sample Values                                                                                                                                                                                                                                                                                 | Remarks |
|-------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| List of Payment details |            | [<br>{<br>"id": 1931,<br>"subscriptionId": 953,<br>"dueDate": "2021-07-14",<br>"status": "SUCCEEDED_PAYMENT",<br>"trxId": "8GE104UBL5",<br>"trxTime": "2021-0714T12:48:22.80292",<br>"amount": 50.00,<br>"reverseTrxAmount": null,<br>"reverseTrxId": null,<br>"reverseTrxTime": null<br>}, ] |         |

## RESPONSE EXAMPLE:

```
[
{
  "id": 1931,
  "subscriptionId": 953,
  "dueDate": "2021-07-14",
  "status": "SUCCEEDED_PAYMENT",

  "trxId": "8GE104UBL5",
  "trxTime": "2021-07-14T12:48:22.80292",
  "amount": 50.00,
  "reverseTrxAmount": null,
  "reverseTrxId": null,
  "reverseTrxTime": null
},
{
  "id": 1932,
  "subscriptionId": 953,
  "dueDate": "2021-07-29",
  "status": "SUCCEEDED_PAYMENT",
  "trxId": "8GT904USN9",
  "trxTime": "2021-07-29T06:00:09.139887",
  "amount": 50.00,
  "reverseTrxAmount": null,
  "reverseTrxId": null,
  "reverseTrxTime": null
}
],
```

## Cancel Subscription using subscription ID

Merchant can cancel a subscription using subscription ID if it needs to. Cancel subscription means there will be no payment charging for the next cycle.

| METHOD        | URL                                                       |
|---------------|-----------------------------------------------------------|
| <b>DELETE</b> | <b>/api/subscriptions/{subscriptionId}?reason={value}</b> |

| TYPE        | Data Type | PARAMETERS            | SAMPLE VALUE     | Remarks                                                                        |
|-------------|-----------|-----------------------|------------------|--------------------------------------------------------------------------------|
| <b>HEAD</b> | String    | Version               | V1.0             | The version number is used to control the system version of both parties.      |
| <b>HEAD</b> | Number    | channelId             | <MERCHAT WEB>    | Channel Id indicates channel by which user initiated the request.              |
|             |           |                       |                  | 1: Customer APP<br>2: Merchant WEB                                             |
| <b>HEAD</b> | Instant   | Timestamp             |                  | Will be used to combat replay attack.                                          |
| <b>HEAD</b> | String    | Content-Type          | application/json | Content Type of request parameters                                             |
| <b>PATH</b> | Number    | subscriptionId        |                  | Subscription ID which can be fetched via execute subscription API              |
| <b>PATH</b> | String    | reasonForCancellation |                  | Defines the reason for cancellation the subscription,<br><b>Optional value</b> |
| <b>HEAD</b> | String    | X-API-KEY             | <string>         | API key to maintain authentication between bKash and merchant servers          |

Response: JSON

| Type   | Data Type | Parameters            | Values                      | Remarks                                                        |
|--------|-----------|-----------------------|-----------------------------|----------------------------------------------------------------|
| ACCEPT |           | <b>Content-Type</b>   | application/json            |                                                                |
| JSON   | String    | subscriptionRequestId | < TraggSB00108>             | Subscription Request ID as identifier of specific request      |
| JSON   | Number    | subscriptionId        | <141>                       | Subscription ID as identifier of specific subscription         |
| JSON   | String    | subscriptionStatus    | <CANCELLED>                 | Subscription status on current time                            |
| JSON   | String    | Message               | <string>                    | An informal description that provides context for transactions |
| JSON   | Instant   | Timestamp             | 2021-07-14T11:27:34.462238Z |                                                                |

RESPONSE EXAMPLE:

```
{
  "message": null
  "timeStamp": "2021-07-14T11:27:34.462238Z",
  "subscriptionRequestId": "TraggSB00108",
  "subscriptionId": 141,
  "subscriptionStatus": "CANCELLED",
}
```

## Subscription Refund

Merchant can refund against a subscription using payment ID if it needs to.

| METHOD      | URL                                     |
|-------------|-----------------------------------------|
| <b>POST</b> | <b>/api/subscription/payment/refund</b> |

| TYPE        | Data Type | PARAMETERS  | SAMPLE VALUE                 | Remarks                                                                                                                                                                             |
|-------------|-----------|-------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>HEAD</b> | String    | Version     | V1.0                         | The version number is used to control the system version of both parties.                                                                                                           |
| <b>HEAD</b> | Number    | channelCode | < Merchant WEB >             | Channel Id indicates channel by which user initiated the request.<br>1: Customer APP<br>2: Merchant WEB                                                                             |
| <b>HEAD</b> | String    | requestId   | <AMQR20200130220330999A1Rtg> | (only alpha numeric no special character) This Id will be used to handling duplicate & end to end message tracing & troubleshooting. This should be unique for the system lifecycle |
| <b>HEAD</b> | Instant   | Timestamp   |                              | Will be used to combat replay attack.                                                                                                                                               |

|      |        |              |                  |                                                                       |
|------|--------|--------------|------------------|-----------------------------------------------------------------------|
| HEAD | String | Content-Type | application/json | Content Type of request parameters                                    |
|      |        |              |                  |                                                                       |
|      |        |              |                  |                                                                       |
| HEAD | String | X-API-KEY    | <string>         | API key to maintain authentication between bKash and merchant servers |

#### Request Body

| Data Type | Parameter | Mandatory /Optional | Sample Value | Remarks                                | Description                                     |
|-----------|-----------|---------------------|--------------|----------------------------------------|-------------------------------------------------|
| Number    | paymentId | M                   | <156778>     | Payment ID received during create call | Payment ID for that Subscription to be refunded |
| Decimal   | amount    | M                   | <120.00>     | Amount to refund                       | Payment Refund Amount                           |

#### RESPONSE:

For Refund API, Merchant will only receive HTTP Status as OK. And after that, merchant will get the Webhook for Refund (success/fail)

**Note:**

- X-API-KEY should be use for all kind of API calls for subscription.
- This X-API-KEY will be provided by bKash once. If merchant requires to change, they can acknowledge bKash to change.
- This is not the concrete API documentation. Few changes might come on the basis of need to improve the API engine.

## Errors

Standard REST API HTTP status code is maintained over all API's related with Subscription Payment.

For all API 200 status code means intended action has been executed successfully.

List of HTTP status are written below:

| internal_code                   | external_code | Description                                                                 |
|---------------------------------|---------------|-----------------------------------------------------------------------------|
| METHOD_ARGUMENT_INVALID         | 1001          | Invalid method argument                                                     |
| METHOD_ARGUMENT_TYPE_MISMATCH   | 1002          | %s is not a valid %s                                                        |
| HTTP_MESSAGE_UNREADABLE         | 1003          | Http message not readable                                                   |
| DATA_INTEGRITY_VIOLATION        | 1004          | Some key violates integrity constraint                                      |
| INVALID_SUBSCRIPTION_FIELD      | 2000          | Invalid %s                                                                  |
| START_DATE_EXCEEDS_CURRENT_DATE | 2001          | Subscription with current date as Start date after 11:30 pm is not allowed. |
| START_DATE_IS_PREVIOUS          | 2002          | Start date cannot be previous date                                          |
| START_DATE_AFTER_EXPIRY         | 2003          | Start date cannot be after expiry date                                      |

|                                    |      |                                                  |
|------------------------------------|------|--------------------------------------------------|
| SUBSCRIPTION_REQUEST_ID_USED       | 2004 | subscriptionRequestId already used               |
| SERVICE_ID_NOT_FOUND               | 2005 | ServiceId %s not found.                          |
| SERVICE_ID_DISABLED                | 2006 | ServiceId %s is disabled                         |
| SERVICE_ID_FOR_REQUESTER_DISABLED  | 2007 | Service id %s with this requester %s is disabled |
| SERVICE_ID_FOR_REQUESTER_NOT_FOUND | 2008 | Service id %s with this requester %s not found   |
| SUBSCRIPTION_NOT_FOUND             | 2009 | Subscription with id: %s doesn't exist           |
| SUBSCRIPTION_NOT_CANCELLABLE       | 2010 | Subscription with id: %s is not cancellable      |
| MERCHANT_NOT_VALID_FOR_REQUESTER   | 2011 | Merchant %s is not valid for requester %s        |
| MERCHANT_NOT_FOUND_SHORTCODE       | 2012 | Merchant %s not found                            |
| REQUESTER_NOT_FOUND_WITH_API_KEY   | 2013 | Requester not found for api key %s               |
| REQUESTER_NOT_AUTHORIZED           | 2014 | Unauthorized to perform action on %s : %s        |
| SUBSCRIPTION_NOT_EXTENDABLE        | 2015 | Subscription with id: %s is not extendable       |
| EXTENDED_END_DATE_INVALID          | 2016 | Extended end date invalid                        |
| SERVICE_ID_INVALID                 | 2017 | Service id is not valid                          |

|                                             |      |                                                  |
|---------------------------------------------|------|--------------------------------------------------|
| TRANSACTION_AMOUNT_BELLOW_CONFIGURED_LIMIT  | 2018 | %s should be greater than configured limit       |
| MERCHANT_DISABLED                           | 2019 | Merchant: %s is disabled                         |
| MERCHANT_NOT_LINKED                         | 2020 | Merchant %s is not linked                        |
| START_DATE_OR_EXPIRY_DATE_INVALID           | 2021 | Start date or expiry date is missing or invalid  |
| START_DATE_AND_EXPIRY_DATE_SAME             | 2022 | Same start date and expiry date is not supported |
| EXPIRY_DATE_BELLOW_THRESHOLD                | 2023 | Expiry date invalid for given frequency          |
| MERCHANT_NOT_FOUND_WITH_ID                  | 3000 | Merchant not found for id : %s                   |
| FIELD_VALUE_INVALID                         | 3001 | %s is not a valid %s                             |
| PAYMENT_NOT_FOUND_WITH_ID                   | 3003 | Payment not found for id: %s                     |
| SUBSCRIPTION_NOT_FOUND_WITH_ID              | 3004 | Subscription not found for id : %s               |
| SUBSCRIPTION_NOT_CANCELLABLE_ADMIN          | 3005 | Subscription with id: %s is not cancellable      |
| MERCHANT_PASSWORD_ENCRYPTION_FAILED         | 3006 | Merchant password encryption failed              |
| MERCHANT_WITH_WALLET_SHORTCODE_EXISTS       | 3007 | Merchant with Wallet or ShortCode exists         |
| OTHER_MERCHANT_WITH_WALLET_SHORTCODE_EXISTS | 3008 | Another Merchant with Wallet or ShortCode exists |

|                         |      |                                                                     |
|-------------------------|------|---------------------------------------------------------------------|
| SERVICE_ID_EXISTS       | 3009 | Service with %s : %s exists                                         |
| SERVICE_WITH_API_SCOPE  | 3010 | Another service with given API Scope and CommandId Exists           |
| INVALID_PASSWORD_LENGTH | 3011 | Invalid password length                                             |
| INVALID_SMS_LENGTH      | 3013 | Length of %s should be less than %s characters without placeholders |
| MERCHANT_EXISTS         | 3015 | Merchant with name or identifier exists                             |